Self-Heterogeneous Knowledge Fusion With Personalized Guidance for Target Fine-Grained Classification in Remote Sensing Images

Hao Zhou, Tingjin Luo, *Member, IEEE*, Yongming He, and Wenxia Guo

***Abstract*—Fine-grained visual classification in remote sensing (FGVC-RS) images is a critical task with broad applications in military reconnaissance, traffic monitoring, and ecological investigation. Existing approaches predominantly adopt a "feature fusion + single classifier" paradigm, where multiple semantic features are pooled or concatenated into a unified representation. However, this often leads to insufficient feature utilization and loss of discriminative details, limiting the model's ability to handle high interclass similarity and intraclass variance. To address these limitations, we propose a novel self-heterogeneous knowledge fusion with personalized guidance (SHKF-PG) framework that adopts a "self-heterogeneous knowledge fusion (SHKF) + multiexpert ensemble" architecture to better leverage multilevel semantic features. Specifically, to focus on extracting discriminative information from unique feature combinations, our SHKF network integrates shallow local, middle-layer attention, and deep global features to form distinct heterogeneous representations, each directed to a specialized expert classifier. To prevent collaborative errors among experts, we introduce a personalized guidance (PG) mechanism that dynamically suppresses the hardest negative class for each expert, steering the model toward correct optimization. Finally, compared with the existing state-of-the-art (SOTA) methods, the extensive experimental results on three benchmarks FGSC-23, FGSCR-42, and Aircraft-16 demonstrate the effectiveness and superiority of our proposed SHKF-PG. Our code will be made publicly available at https://github.com/zhouhao0515/SHKF-PG.**

***Index Terms*—Ensemble learning, fine-grained visual classification, personalized guidance (PG), remote sensing, self-heterogeneous knowledge fusion (SHKF).**

I. Introduction

FINE-GRAINED visual classification in remote sensing (FGVC-RS) images is a core and challenging task within the field of remote sensing image interpretation [1], [2]. It aims to distinguish fine-grained subordinate object categories, such as differentiating between the Arleigh Burke class destroyer and the Akizuki class destroyer. With the rapid development of high-resolution remote sensing satellites, FGVC-RS has demonstrated significant application potential in military reconnaissance, traffic monitoring, and ecological investigation [3], [4]. However, unlike fine-grained classification in natural images [5], [6], FGVC-RS presents unique challenges: high intraclass variance caused by variations in imaging viewpoint, illumination, occlusion, and other factors, and high interclass similarity, such as the visual resemblance between different types of destroyers [7], [8], as illustrated in Fig. 1(a) and (b). These challenges make them more difficult for traditional classification methods to learn highly discriminative feature representations. Moreover, the requirement for high classification accuracy imposes stringent demands on both feature representation and decision-making mechanisms.

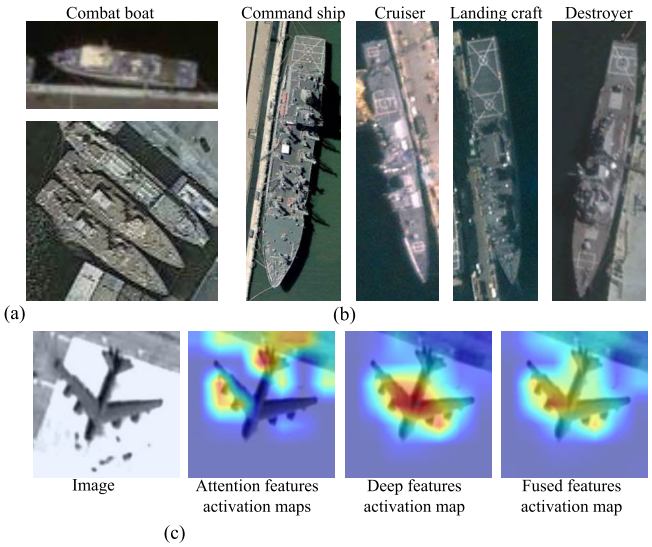


Fig. 1. Challenges of FGVC-RS. (a) High intraclass variance, (b) high interclass similarity, and (c) fusing multiple semantic features into a single global representation might lead to information loss in the "feature fusion + single classifier" paradigm.

In recent years, deep convolutional neural networks (DCNNs) have achieved remarkable progress in fine-grained classification tasks [9]. To overcome the challenges of high interclass similarity and high intraclass diversity, current research in FGVC-RS has primarily focused on feature learn-

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ing [10] and contrastive learning [7]. In feature learning, existing models extract rich and discriminative multilevel features through local feature enhancement and hierarchical information integration. For instance, Yang et al. [11] proposed the mid-level feature causal attention (MFCA) model. It constrained mid-level feature learning by comparing attention maps with counterfactual attention maps, forcing the model to focus on discriminative regions such as ship decks. Several studies have indicated that shallow or middle-layer features in networks contain richer discriminative information. The edge and texture information also has predictive power for certain fine-grained samples. For example, Zhang et al. [12] leveraged self-distillation techniques to fully exploit the value of shallow features. Jin et al. [13] also pointed out that different classes have different preferences to be fit by models with different depths. Yi et al. [10] introduced a multiple hierarchies attention module (MHAM) in the EFM-Net model to extract features from different network layers and filter highly discriminative local patterns through channel attention. Li and Bian [14] proposed a feature reconstruction network (FRN), which calculated weights for different feature channels via a ridge regression algorithm, thereby suppressing background interference through foreground weighting. Additionally, metric optimization methods [15] based on contrastive learning aim to improve interclass separability by designing discriminative metrics. Chen et al. [7] proposed the push-and-pull network (P2Net). It first separated image features via contrastive learning and then aggregated samples of the same class using proxy vectors for intraclass distribution dispersion. Xiong et al. [16] proposed the cognitive network (Cog-Net), which simulated the human reasoning process from perception to cognition. It extracted global–local features through causal multigranularity feature learning and then generated interpretable decisions via visual concept learning. These methods have achieved positive progress in FGVC-RS.

However, existing approaches still exhibit limitations in fully leveraging the value information from multilevel features for informed decision-making. A predominant issue lies in the widely adopted "feature fusion + single classifier" paradigm. On the one hand, the feature utilization paradigm is often simplistic and insufficient. Shallow features typically capture edge and texture details, deep features encode global semantic information, and attention features may highlight key parts. Most of the existing methods [10], [11], [17] simply fused multiple semantic features into a single global representation via pooling or concatenation operations before feeding them into a unified classifier. This overlooked the complementary discriminative information in different level features and failed to provide independent and sufficient learning in decision-making space for such information. On the other hand, such fusion strategies might lead to information loss. As shown in Fig. 1(c), some operations like bilinear pooling and global average pooling smooth out feature details in feature fusion. EFM-Net [10] captured global–local correlations through bilinear polymerization pooling, but it ignored the unique contribution of individual elements within the feature vector. For example, the mast shape of a specific ship might be diminished after pooling. These limitations restrict the model's ability to precisely delineate decision boundaries and enhance feature space discriminability when facing FGVC-RS tasks with high interclass similarity and high intraclass variance. Therefore, both extracting diverse semantic features from images and improving the utilization of information within these features are crucial for advancing FGVC-RS. Yet, existing models typically focus on the former while neglecting the latter.

To tackle the above problems, we propose a novel self-heterogeneous knowledge fusion with personalized guidance (SHKF-PG) model for fine-grained remote sensing image classification. Departing from the conventional "feature fusion + single classifier" architecture, SHKF-PG introduces a novel "self-heterogeneous knowledge fusion (SHKF) + multiexpert ensemble" learning framework. This design assigns different hierarchical features to distinct experts, enabling them to cultivate specialized, heterogeneous knowledge from the same backbone network. Consequently, the model maximizes the utilization of discriminative information embedded across semantic features, thereby mitigating information loss and promoting feature complementarity. Notably, a critical risk in such a design is that different experts might converge on incorrect predictions regarding hard negative classes, thereby misleading the entire model. To mitigate this, we also design a personalized guidance (PG) mechanism to suppress consensus on the hardest negative classes among experts and guide the model toward correct learning directions. Specifically, SHKF-PG comprises three core components: a multilevel feature extraction module, an SHKF network, and a PG module. The feature extraction module hierarchically captures diverse semantic features—from shallow local details to middle attention maps and deep global representations—enabling comprehensive scene understanding. The SHKF network integrates this heterogeneous knowledge by fusing shallow and attention features into the deep global features of each expert, enhancing expert diversity and feature enrichment. Mutual knowledge distillation (MKD) is employed to facilitate information exchange among experts, fully exploiting the diversity of the mixture-of-experts (MoE) architecture for optimized performance. The PG module constructs a guidance teacher from ensemble predictions, targeting each expert's specific biases (e.g., overconfidence in hard negatives) to reinforce ground-truth signals and suppress erroneous consensus. During inference, all expert classifiers collaborate to generate final predictions, balancing individual strengths and compensating for weaknesses. Finally, extensive experiments on the FGSC-23, FGSCR-42, and Aircraft-16 datasets demonstrate the superiority of our model. Compared to other state-of-the-art (SOTA) models, the SHKF-PG model improves the overall accuracy (OA) by 1.4%, 0.2%, and 0.6%. In summary, the main contributions are summarized as follows.

1) We propose a novel "SHKF + multiexpert ensemble" framework for fine-grained remote sensing image classification. Our approach enhances decision diversity and discriminative power by leveraging dedicated experts to exploit distinctive hierarchical representations.
2) We introduce an SHKF network that enables each expert to extract valuable information from multilevel seman-

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tic features for decision-making, which significantly improves the utilization efficiency of heterogeneous semantic information.

3) We design a PG mechanism that dynamically identifies and suppresses the hardest negative class across experts while providing tailored feedback to individual experts, which mitigates error propagation and steers the model toward correct optimization directions.

II. Related Works

A. Feature Learning-Based Fine-Grained Visual Classification Models

Feature learning-based methods aim to extract discriminative features from remote sensing images to achieve fine-grained classification. Current research primarily focuses on attention mechanisms and feature fusion to help networks identify and localize discriminative object features [10], [18], [19]. Attention-based models locate key parts of targets and suppress background interference through specialized modules [11], [20]. Sermanet et al. [21] first introduced the attention mechanism into fine-grained classification tasks, which effectively improved model performance. Yang et al. [22] proposed an anchor-guided region-level localization model, and they used attention feature maps to accurately locate local regions. Jiang et al. [23] proposed MP-FGVC, a CLIP-based multimodal prompting method that leverages subcategory-specific vision prompts and discrepancy-aware text prompts for fine-grained visual classification. Wang et al. [24] employed a self-attention mechanism to enhance global feature representations. Li et al. [25] used attention to compute channel-wise weights for feature reconstruction, improving few-shot fine-grained classification. Causal learning has also been applied to extract discriminative features [26], [27]. Xiong et al. [17] proposed the CMAM model, which explores the causal relationship between multiple discriminative local regions and prediction results through a multihead attention mechanism.

For feature fusion, many studies have shown that shallow and middle-layer features contain rich discriminative information, often combined with deep features for classification [10], [11], [28]. Du et al. [29] extracted multigranularity feature maps and used attention to fuse and classify them, improving multiscale object classification. Zhang et al. [30] introduced a multilevel enhanced feature representation method that integrates global and local features for fine-grained ship classification. Yang et al. [11] proposed a mid-level feature attention learning method to fuse causal attention features and channel attention features to identify discriminative regions. Yi et al. [10] proposed the multiple hierarchies attention in EFM-Net, which extracts and fuses features from different network layers via attention to capture essential object features.

B. Contrastive Learning-Based Fine-Grained Visual Classification Models

Contrastive learning has recently emerged as a research topic in unsupervised and self-supervised representation learning [31], [32]. It aims to learn discriminative features by contrasting positive and negative samples [33]. In fine-grained visual classification, metric learning and contrastive learning can help models better capture intraclass similarities and interclass differences [7], [15]. Liu et al. [34] proposed an attentive contrastive learning network (ACLN) with two contrastive losses to attract positive pairs and repel negative pairs. Kim and Ha [35] proposed the C3-GAN model, which utilized a discriminator for both classification and clustering. The C3-GAN model optimized a contrastive loss to maximize mutual information between positive samples while minimizing it between negative samples. To capture subtle local differences in complex remote sensing scenes, Li et al. [36] designed a global–local contrastive learning strategy. By employing contrastive tasks at both global and local granularities, they effectively captured contrastive clues at multiple levels. Chen et al. [7] proposed the push-and-pull network (P2Net), which forces all instances to be uncorrelated and as far away as possible in the push-out stage, and groups instances into each subclass during the pull-in phase. Xiong et al. [16] introduced visual concept learning in Cog-Net, imitating the human recognition process to learn basic visual concepts for each ship category. Xu and Lang [15] proposed a metric learning framework to improve the performance of ship classification in remote sensing images. They embedded a distribution loss branch in the UMP+D model to refine the sample distribution, further tightening intraclass samples and pulling apart interclass samples.

C. Multiexpert Ensemble Learning

Multiexpert ensemble learning combines multiple expert classifiers, which has achieved remarkable performance in image classification [37], [38]. Classical MoE models [39] were inherently sample-oriented, as they learned optimal routing strategies to allocate samples to appropriate experts. While effective for tasks emphasizing the global distribution patterns, they falled short in capturing subtle local differences for fine-grained classification. For imbalanced image classification, a common practice is to divide the dataset into different subsets to train distinct experts. Cai et al. [40] divided the dataset into different overlapping subsets and equipped each subset with an expert classifier to learn supportive and complementary knowledge. Similarly, Zhou et al. [37] designed heterogeneous knowledge transfer between experts trained on overlapping subsets. In fine-grained visual classification, multiexpert ensemble learning can better handle different types of samples through the collaboration of multiple expert models [2], [41], [42]. However, it is proven that dividing the dataset for training different experts is often infeasible in fine-grained classification due to limited data [2]. Therefore, Zhang et al. [2] used visual attention to provide experts with full-size data at different granularities, enhancing diversity. Xu et al. [41] proposed the IELT model for fine-grained classification tasks, integrating the idea of ensemble learning into vision transformers. They designed a multihead voting (MHV) module that utilizes attention maps and spatial relationships to select discriminative regions from each layer. Zhang et al. [42] proposed a CLIP-based MoE (CLIP-MoE) model, which is a fine-tune fine-grained framework built upon large-scale cross-modal pretraining. While leveraging pretraining benefits,

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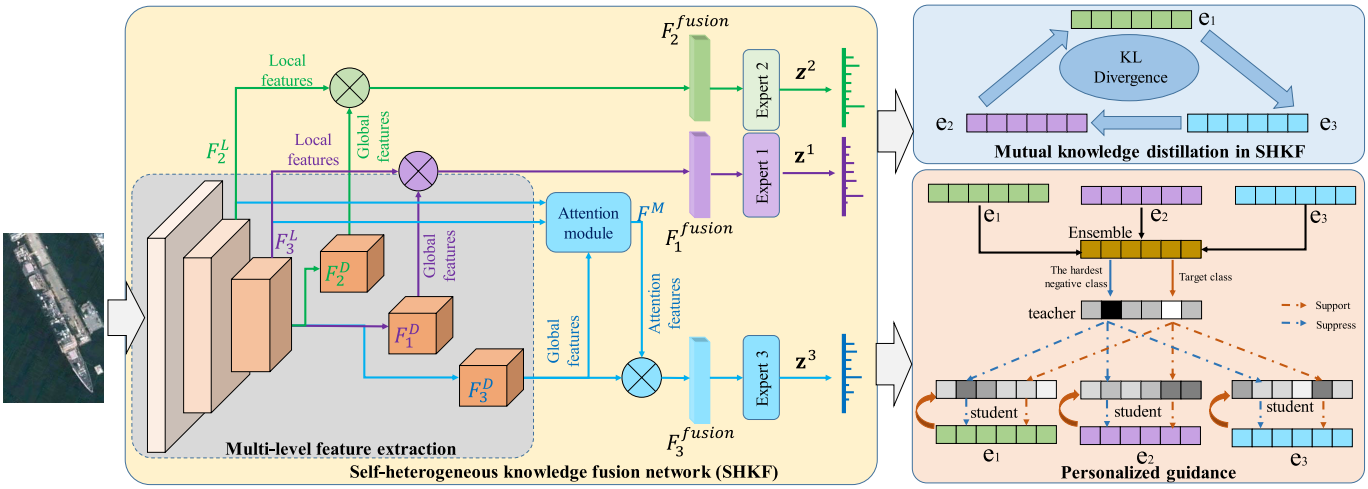


Fig. 2. Main pipeline and structure of our proposed SHKF-PG. It consists of three core components: a multilevel feature extraction module to extract multilayer semantic features, an SHKF network to excavate valuable information and heterogeneous knowledge with MKD, and a PG module to suppress hard negative classes for correct classifications.

CLIP-MoE did not specialize in remote sensing image and lack mechanisms to handle specific challenges. Shi et al. [43] proposed a multifeature ensemble model to extract global and local discriminative features for fine-grained ship recognition in remote sensing. They used preprocessing methods to extract multiple features as input to a CNN and equipped each feature with an expert classifier.

III. Method

A. Model Framework

We propose the SHKF-PG model, a novel "SHKF + multiexpert ensemble" architecture designed for fine-grained classification in remote sensing images. As illustrated in Fig. 2, the SHKF-PG model comprises three key components: a multilevel feature extraction module, an SHKF network, and a PG module. The multilevel feature extraction module extracts discriminative semantic features from input images, including shallow local features, middle-layer attention features, and deep global features. These multigranularity representations serve as the basis for subsequent decision-making. To fully leverage these features, we design an SHKF network incorporating multiple expert networks. Each expert fuses shallow and middle-level features with deep global features to enhance representational diversity. To facilitate collaboration among different experts, MKD is employed within the fusion network to enable effective message passing. Furthermore, to mitigate the risk of consensus errors among experts on hard negative samples, we introduce a PG module. This component provides tailored supervision to each expert by dynamically suppressing the hardest negative class based on ensemble predictions, thereby steering the model toward correct classifications.

B. Preliminaries

Let the proposed SHKF-PG model, parameterized by θ, consist of M experts based on a ConvNeXt backbone [44]. Typically, the multiexpert network shares the initial layers across all experts, while the final layers are exclusive to each expert. In the SHKF-PG model, the first $S-1$ stages of the backbone are shared. Only the final stage is employed as the exclusive parameter for each expert. Without loss of generality, for expert m, we denote its exclusive parameters as θ_{S_m}. Given an input image I, the shallow local features F_s^L from the shared layers at stage s are expressed as

$$F_s^L = \theta_s \circ \theta_{s-1} \circ \cdots \circ \theta_1 (I) \tag{1}$$

where $\circ$ denotes function composition: $h \circ g(x) = h(g(x))$. After the shared network, the deep global features F_m^D output from the final stage for the mth expert can be expressed as

$$F_m^D = \theta_{S_m} \left(F_{S-1}^L \right). \tag{2}$$

In this work, we use the ConvNeXt architecture with $S = 4$ stages.

C. Multilevel Feature Extraction Module

Existing works often focus on extracting multilevel features or attention features to obtain rich and discriminative representations. In the SHKF-PG model, we design a multilevel feature extraction module to capture shallow local features, middle-layer attention features, and deep global features as the cornerstone for decision-making. These features provide comprehensive support for the model's final predictions.

The multilevel feature extraction module employs a ConvNeXt backbone due to its simple and efficient block design, extracting both shallow local and deep global features. We extract the shallow local features from the first s stages $F_1^L, \ldots, F_s^L$, which provide essential edge and texture information, and deep global features F_m^D from the final stage, which supply high-level semantic information for the mth expert. Middle-layer features often contain richer discriminative information and are widely used in fine-grained recognition. To extract discriminative object part features, we employ multiple middle-layer features as input to generate middle-layer attention features F^M via multiple hierarchies attention mechanism (MHAM) [10]. MHAM is a dual-path architecture composed

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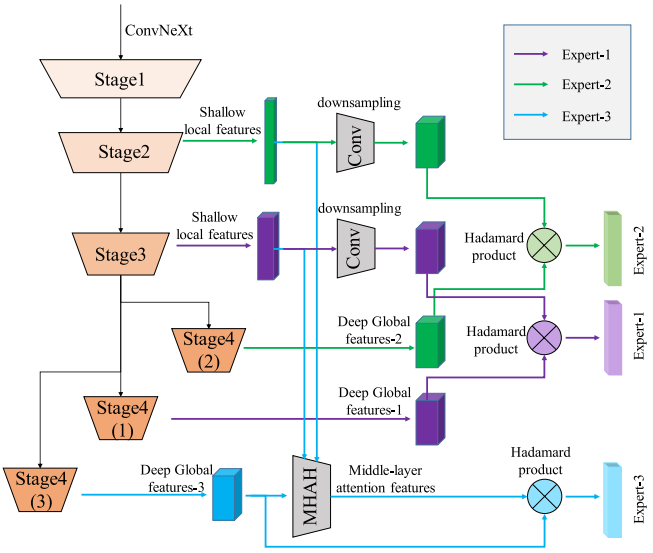


Fig. 3. Structure of the SHKF network.

of multiple attention blocks designed to identify the most recognizable features. The right-to-left pathway propagates high-level features through upsampling for feature fusion, while the left-to-right pathway transfers texture and edge information. Specifically, we use the outputs from the last three stages of the ConvNeXt network as input for MHAM. The middle-layer attention features F^M can be formulated as

$$F^M = \text{MHAM}\left(F^L_{S-2}, F^L_{S-1}, F^D_M\right) \tag{3}$$

where MHAM$(\cdot)$ is the multiple-hierarchy attention module.

D. SHKF Network

Different levels of features capture distinct information: shallow layers capture local edges, mid-level layers focus on object part information, and deep layers encompass global semantics. To leverage multilevel semantic features more effectively, we propose an SHKF network based on a multiexpert architecture for heterogeneous knowledge learning. As shown in Fig. 3, our design principle is to specialize experts via "division of labor" at the feature level. Each expert takes deep global features along with different shallow local or middle-layer attention features as input to exploit valuable information comprehensively. Through multiexpert ensemble learning, heterogeneous hierarchical features from the same backbone network will form SHKF.

Formally, we assign the $m-1$ shallow local features and the middle-layer attention features F^M to m experts, fusing them with each expert's deep global feature. Specifically, for the kth expert ($k = 1, \ldots, m-1$), it receives the shallow feature F^L_{S-k} from stage $S-k$ fused with its own deep global feature F^D_k. If $m > S$, we cycle back through the shallow stages, e.g., stage $S-1$ is reused for the Sth expert, to ensure that each of the first $m-1$ experts gets a distinct shallow feature. The mth expert receives the middle-layer attention feature F^M generated by MHAM fused with its deep global feature F^D_m. Since the assigned features have different spatial dimensions, we apply several convolutional layers for downsampling to achieve feature alignment. For the nth expert, the aligned shallow local features $\hat{F}^L_{S-n}$ are obtained by

$$\hat{F}^L_{S-n} = \text{downsampling}\left(F^L_{S-n}\right) \tag{4}$$

where downsampling$(,)$ consists of several convolutional layers. The aligned features are then fused with deep global features F^D_n via Hadamard product to form the fused features F^{fusion}_n

$$F^{fusion}_n = \hat{F}^L_{S-n} \otimes F^D_n \tag{5}$$

where $\otimes$ is the Hadamard product. The fused features F^{fusion}_n are then passed through a linear layer to produce the logits $\mathbf{z}^n$ for expert n and represented as

$$\mathbf{z}^n = W_n \cdot F^{fusion}_n + b_n \tag{6}$$

where W_n and b_n are the weight matrix and bias of the linear layer for expert n, respectively.

Similarly, the middle-layer attention features F^M are fused with the deep global features F^D_m of the mth expert after downsampling, producing the logits $\mathbf{z}^m$

$$\hat{F}^M = \text{downsampling}\left(F^M\right) \tag{7}$$

$$F^{fusion}_m = \hat{F}^M \otimes F^D_m \tag{8}$$

$$\mathbf{z}^m = W_m \cdot F^{\text{fusion}}_m + b_m \tag{9}$$

where $\hat{F}^M$ represents the aligned middle-layer attention features and W_m and b_m are the corresponding weight matrix and bias. Through the SHKF network, the shallow local features and middle-layer attention features are assigned to dedicated experts, respectively.

During training, the cross-entropy loss L_i is computed for the logits of each expert i. The overall multiexpert loss L_{ME} is defined as

$$L_{ME} = \frac{1}{m}\left(\sum_{i=1}^{m} L_i\right). \tag{10}$$

To facilitate collaboration among experts and integration of knowledge from different features, we employ MKD between experts to make them learn from each other. Since each expert aggregates different features, both of them can act as either a teacher or a student. We employ the KL divergence to measure the distribution differences between experts and minimize it to achieve message passing between experts. Thus, the MKD loss among m experts can be expressed as

$$L_{MU} = \sum_{j=1}^{m}\sum_{k\neq j}^{m} KL\left(P^j \| P^k\right) \tag{11}$$

where $KL(\|)$ denotes the function of KL divergence and P denotes the probability distribution obtained by applying the softmax function to the logits from experts.

Through SHKF and MKD, the model effectively integrates edge features, part features, and global semantic features from different network depths. It fully leverages valuable information from diverse semantic features to support the final decision. The aggregation of information during mutual distillation further promotes collaboration among experts, enhancing the effectiveness of multiexpert ensemble learning. In our implementation, we set $m = 3$.

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E. PG Mechanism

In image classification, nontarget classes with high prediction scores are often referred to as hard negative classes [13], [45]. Multiexpert networks require each expert to contribute unique and complementary information. It is critical to avoid consensus among experts on the hardest negative class for a given sample, as such consensus may propagate and reinforce erroneous information during knowledge transfer, ultimately misleading the model. General ensemble regularization methods primarily aim to improve generalization and prevent overfitting via weight and parameter regularization. However, these methods force all experts to converge toward a unified output through generalized constraints, which cannot solve the challenge of the hardest negative class.

To address this issue, we propose a novel PG mechanism to dynamically suppress the hardest negative class in multiexpert classification. Its core idea is "teaching according to aptitude," providing tailored guidance to each expert based on its specific weaknesses or biases. In fine-grained remote sensing image classification, different experts may develop unique classification biases due to their focus on different features. For example, an expert focusing on shallow local features may underestimate the target class confidence, while an expert focusing on middle attention features may overestimate a similar negative class. A unified teacher guidance struggles to accommodate these differences, whereas PG allows each expert to address its shortcomings while preserving its strengths. To preserve expert diversity, the PG mechanism does not eliminate the differences among experts but guides them toward complementary behavior. Therefore, the teacher logits in the PG mechanism are derived from the collective wisdom of all experts, aiming to amplify the target class signal and suppress the consensus negative class with the highest confidence.

1) Identifying the Hardest Negative Class From Multiexpert Predictions: For a sample x with its ground-truth one-hot vector $\mathbf{y} = (y_1, y_2, \ldots, y_C)$, where C is the number of categories, we denote that the output logits of expert n are $\mathbf{z}^n = \{z_1^n, z_2^n, \ldots, z_C^n\}$. The average logits over all experts for each class can be computed as $\bar{\mathbf{z}} = \{\bar{z}_1, \bar{z}_2, \ldots, \bar{z}_C\}$, where

$$\bar{z}_i = \frac{1}{m} \sum_{j=1}^{m} z_i^j. \tag{12}$$

Therefore, the hardest negative class is identified as the class i with the maximum average logit value $\max_i\{\bar{z}_i\}$ among nontarget classes ($i \neq j$ and $y_j = 1$).

2) Generating PG Teacher: To preserve expert diversity, we do not directly adjust the output logits of each expert in the PG mechanism. Instead, we provide directional guidance based on the differences between each expert's logits and the multiexpert average logits. By calculating these differences, we can more effectively identify the weaknesses of each expert, thereby suppressing the hardest negative classes while encouraging them to make correct classifications. As shown in Fig. 4, for expert n, the difference vector between its logits $\mathbf{z}^n$ and the ensemble logits $\bar{\mathbf{z}}$ is defined as $\hat{\mathbf{z}}^n = \{\hat{z}_1^n, \hat{z}_2^n, \ldots, \hat{z}_C^n\}$,

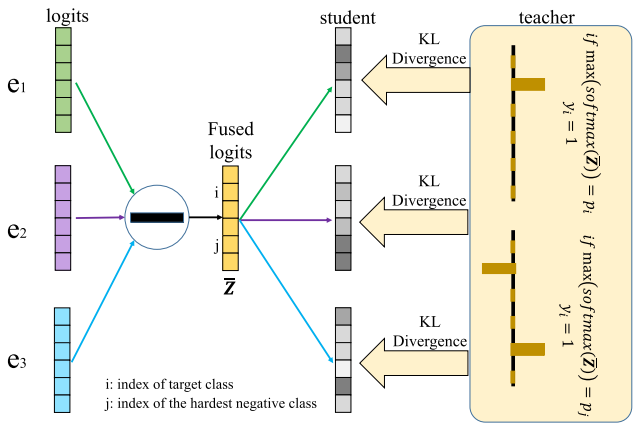


Fig. 4. PG loss calculates the KL divergence.

where

$$\hat{z}_i^n = z_i^n - \bar{z}_i. \tag{13}$$

The difference vector $\hat{\mathbf{z}}^n$ reflects the relative strengths and weaknesses of each expert in ensemble decision-making. For instance, when an expert's logit value for the ground-truth class is lower than others, a negative value in the difference vector indicates that the expert should increase its confidence. Conversely, if an expert's logit value for the hardest negative class exceeds others, a positive value in the difference vector indicates an overestimation that needs correction.

Therefore, to suppress the hardest negative class during misclassification and encourage the ground-truth item in logits, we design a guidance teacher based on the difference vector to provide personalized supervision for each expert. The purpose of PG is to make the ground-truth item in the difference vector positive and the hardest negative item negative. Specifically, we define a standardized guidance vector as the teacher to direct the model's adjustments, primarily indicating whether predictions should be reinforced or suppressed. When the model classifies correctly, the guidance teacher provides positive feedback, encouraging experts to further increase confidence in the correct class. When the model misclassifies, the teacher provides negative feedback, suppressing the hardest negative class while steering experts toward the correct class. The training process is then regularized by minimizing the KL divergence between the difference vector and the guidance teacher.

Specifically, we denote the estimated probability distribution of the SHKF-PG model $\mathbf{p} = (p_1, p_2, \ldots, p_C) = \text{softmax}(\bar{\mathbf{z}})$ and the ground-truth one-hot vector $\mathbf{y} = (y_1, y_2, \ldots, y_C)$. If $\max(\mathbf{p}) = p_i$ and $y_j = 1$, then the PG vector $\mathbf{z}^{gv}$ can be expressed as

$$\mathbf{z}^{gv} = \left(z_1^{gv}, z_2^{gv}, \ldots, z_C^{gv}\right) \tag{14}$$

where

$$\begin{cases} z_j^{gv} = 1, z_k^{gv} = 0 & when \;\; i = j \;\; and \;\; k \neq i \\ z_i^{gv} = -1, z_j^{gv} = 1, z_k^{gv} = 0 & when \;\; i \neq j \;\; and \;\; k \neq i. \end{cases} \tag{15}$$

According to (14) and (15), when the multiexpert model correctly classifies a sample ($i = j$), the item corresponding

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to the ground-truth class in guidance teacher is set to 1 ($z_j^{gv} = 1$), thereby providing positive feedback to all experts and encouraging them to further increase the confidence score for the correct class. Conversely, when the model misclassifies ($i \neq j$), the item with the highest prediction score becomes the hardest negative class z_i^{gv}. In this case, the item corresponding to the hardest negative class in guidance teacher is set to -1 ($z_i^{gv} = -1$), delivering negative feedback that forces all experts to suppress the confidence for the hardest negative class. Simultaneously, the item for the ground-truth class remains set to 1 ($z_j^{gv} = 1$), guiding the model to increase the confidence for the correct classification.

3) Computing KL Divergence for PG: We primarily compute the KL divergence between the difference vector and the guidance teacher to guide each expert's adjustments. The PG loss (PG loss) L_{PG} can be expressed as

$$L_{PG} = \frac{1}{m}\sum_{i=1}^{m} KL\left(\text{softmax}\left(\mathbf{z}^{gv}\right) \| \text{softmax}\left(\hat{\mathbf{z}}^{i}\right)\right) \tag{16}$$

where $\mathbf{z}^{gv}$ will be selected based on the results of the ensemble prediction using (14) and (15) and $\mathbf{z}^{gv}$ and $\hat{\mathbf{z}}^{i}$ are converted into probability distributions through the softmax function. Importantly, the model only needs to slightly reduce the confidence of the most confusing class so that it falls below that of the target class, rather than suppressing it to the lowest level. This design encourages sharper decision boundaries without forcibly distorting the learned representations and logits, preserving the model's ability to retain reasonable probabilities for semantically similar classes when they are not the primary source of confusion.

During training, we use the cross-entropy loss for each expert, the MKD loss from the SHKF, and the PG loss to suppress the hardest negative classes. The overall objective function is

$$L_{all} = L_{ME} + \alpha L_{MU} + \beta L_{PG} \tag{17}$$

where α and β are tradeoff hyperparameters.

IV. Experiments

In this section, we evaluate the performance of our SHKF-PG approach in different aspects, including classification comparison with SOTA methods and ablation study.

A. Datasets

Following [10] and [11], to validate the effectiveness of the proposed SHKF-PG model, we conduct extensive experiments on three fine-grained classification datasets for remote sensing images: FGSC-23 [30], FGSCR-42 [46], and Aircraft-16 [10]. FGSC-23 and FGSCR-42 are fine-grained ship classification datasets, while Aircraft-16 focuses on fine-grained aircraft classification. Both FGSC-23 and Aircraft-16 exhibit significant sample variation and suffer from severe scene interference, whereas images in FGSCR-42 are generally clearer.

1) FGSC-23 Dataset: FGSC-23 is a high-resolution optical fine-grained ship dataset for remote sensing classification. It comprises 4080 remote sensing images primarily sourced from Google Earth and Gaofen-1 (GF-1) Satellite, with resolutions ranging from 0.4 to 2 m. FGSC-23 dataset contains 23 categories: including 22 fine-grained ship classes and one nonship category, such as aircraft carrier, destroyer, oil tanker, fishing boat, and so on. Image sizes range from 40×40 to 800×800 pixels. Each target is annotated with a class label, an aspect ratio label, and a distribution direction label. Consistent with [10] and [30], we randomly split all samples into training and test sets with an approximate ratio of 8:2.

2) FGSCR-42 Dataset: FGSCR-42 is also designed for fine-grained optical ship classification. It contains 9320 high-resolution images collected from Google Earth and various public remote sensing image datasets, such as the sliced images from DOTA [58]. FGSCR-42 consists of 42 ship categories, including the Kitty Hawk class aircraft carrier, the Arleigh Burke class destroyer, and so on, which are more finely subdivided than those in FGSC-23. The image sizes range from 50×50 to approximately 500×1500 pixels. Following [17] and [46], half of the images of each category are randomly split for training, while the rest for testing.

3) Aircraft-16 Dataset: Aircraft-16 is designed for fine-grained classification of aircraft targets. It contains 8488 cropped images of aircraft targets, primarily sourced from other public remote sensing object detection datasets, such as RAIAirplane [59], with resolutions ranging from 0.5 to 1.5 m. Aircraft-16 encompasses 16 aircraft categories with diverse backgrounds, including the B-52, E-3, and Boeing 747. We adopt the standard split in [10], where the ratio of the training set to the test set is approximately 8:2.

B. Metrics and Implementation Details

To evaluate the classification performance of the proposed network, we mainly use the OA and the accuracy rate (AR) of each category as the main metrics. OA is defined as the ratio of correctly classified images to the total number of test images, while AR refers to the ratio of correctly classified images within a single class. We also report Params (M), FLOPs (G), training time, FPS, and test speed to assess computational efficiency.

In our SHKF-PG model, we use ConvNext as the backbone for feature extraction. For data preprocessing, the images are first resized to 256×256 by bicubic interpolation, and then, a 224×224 crop is taken either from the original or a horizontally flipped image. During testing, only interpolation and center cropping are required. The learning rate is initialized to 1×10^{-3}, with a batch size of 32. The model is trained using AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay of 1×10^{-8}, and epsilon (eps) set to 1×10^{-8}. In MAHM, the number of channels in the attention block is set to 16. Our model is implemented in PyTorch and trained on an NVIDIA GeForce 3090Ti GPU with 24-GB memory under Ubuntu 18.04.

C. Comparisons With SOTA Methods

In this section, we evaluate the proposed SHKF-PG model on the FGSC-23, FGSCR-42, and Aircraft-16 datasets and

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TABLE I
COMPARISON EXPERIMENTAL RESULTS ON FGSC-23. MODELS WITH * DENOTE THAT THEY ARE REIMPLEMENTED IN OUR EXPERIMENTS

Models	Input Size	OA
VGG-16 (2014) [47]	448 × 448	82.0
SGFM-Net(2025) [48]	-	83.5
DenseNet (2017) [49]	448 × 448	84.0
ResNet (2015) [50]	448 × 448	84.6
TASN (2019) [51]	448 × 448	85.8
MFAL(2024) [11]	-	86.2
SIM (2022) [52]	448 × 448	86.3
DCL (2019) [53]	448 × 448	87.5
FFVT (2021) [24]	384 × 384	87.6
PMG-V2 (2021) [29]	448 × 448	88.1
API-Net (2020) [54]	448 × 448	88.7
P2-Net (2024) [7]	224 × 224	89.0
MSCL-Net(2024) [55]	224 × 224	91.5
EA-Net (2022) [17]	224 × 224	92.0
GLCL*(2025) [36]	224 × 224	92.2
EFM-Net w/o Refiner* (2023) [10]	224 × 224	91.6
EFM-Net* (2023) [10]	224 × 224	92.3
AFF (2024) [56]	-	92.5
MGFE-Net(2025) [57]	224 × 224	92.7
SHKF-PG	224 × 224	**93.0**

TABLE II
COMPARISON EXPERIMENTAL RESULTS ON FGSCR-42

Models	Input Size	OA
DenseNet (2017) [49]	448 × 448	88.7
DCL (2019) [53]	448 × 448	93.0
P2-Net (2024) [7]	224 × 224	93.2
TASN (2019) [51]	448 × 448	93.5
AFF(2024) [56]	-	93.9
EA-Net (2022) [17]	224 × 224	94.1
UMP+D(2024) [15]	224 × 224	95.2
BAT (2020) [60]	448 × 448	96.0
SGFM-Net(2025) [48]	-	96.1
API-Net (2020) [54]	448 × 448	97.6
MFAL(2024) [11]	-	97.6
FFVT (2021) [24]	384 × 384	97.9
SIM (2022) [52]	448 × 448	97.9
PMG-V2 (2021) [29]	448 × 448	98.0
Cog-Net(2024) [16]	224 × 224	98.1
GLCL*(2025) [36]	224 × 224	98.6
MGFE-Net(2025) [57]	224 × 224	99.0
EFM-Net w/o Refiner* (2023) [10]	224 × 224	99.2
EFM-Net* (2023) [10]	224 × 224	99.3
SHKF-PG	224 × 224	**99.4**

TABLE III
COMPARISON EXPERIMENTAL RESULTS ON AIRCRAFT-16

Models	Input Size	OA
BAT (2020) [60]	448 × 448	80.4
AC-Net (2020) [61]	448 × 448	81.5
ResNeXt (2017) [62]	448 × 448	83.4
VGG-16 (2014) [47]	448 × 448	84.5
DenseNet (2017) [49]	448 × 448	84.8
ResNet (2015) [50]	448 × 448	85.0
TASN (2019) [51]	448 × 448	86.3
DCL (2019) [53]	448 × 448	89.0
FFVT (2021) [24]	384 × 384	89.2
SGFM-Net(2025) [48]	-	89.3
SIM (2022) [52]	448 × 448	89.5
API-Net (2020) [54]	448 × 448	89.6
PMG-V2 (2021) [29]	448 × 448	89.7
EFM-Net w/o Refiner* (2023) [10]	224 × 224	92.0
EFM-Net* (2023) [10]	224 × 224	92.2
SHKF-PG	224 × 224	**92.6**

compare its performance against SOTA methods in terms of OA. The compared methods include both general-purpose deep learning models and specialized fine-grained models: VGG-16 [47], ResNet [50], DenseNet [49], ResNeXt [62], DCL [53], TASN [51], API-Net [54], AC-Net [61], PMG-V2 [29], EA-Net [17], BAT [60], FFVT [24], SIM [52], EFM-Net [10], P2-Net [7], Cog-Net [16], MFAL [11], AFF [56], UMP+D [15], MSCL-Net [55], GLCL [36], MGFE-Net [57], and SGFM-Net [48]. Traditional fine-grained dedicated models (e.g., DenseNet, B-CNN, and TASN) often use high-resolution inputs of 448 × 448 to capture more spatial details, while recent efficient models (e.g., EFM-Net, Cog-Net, MSCL-Net, MGFE-Net, and our model) focus on discriminative details effectively at moderate resolutions of 224 × 224, achieving a better balance between performance and efficiency. The comprehensive comparative results are summarized in Tables I–III.

1) Performance on FGSC-23: This experiment evaluates the model's robustness against significant intraclass variation and strong background interference, as ships of the same class in this dataset exhibit substantial appearance changes due to differing imaging perspectives and illumination conditions. As shown in Table I, compared to the latest MFAL, P2-Net, EA-Net, MGFE-Net, and GLCL models, the SHKF-PG model achieved performance improvements of 6.7%, 3.9%, 0.9%, 0.3%, and 0.8%, respectively. Input resolution is usually positively correlated with performance in fine-grained classification tasks. SHKF-PG outperforms many earlier models, such as B-CNN and TASN, while using a lower input resolution. It underscores the advantage of our model that does not rely on high-resolution inputs. Notably, it also yields a consistent improvement over the SOTA EFM-Net model. Our model achieves an OA of 93.0%, representing a 1.4% improvement over EFM-Net under fair conditions without Refiner module. We attribute this superiority to overcoming the limitations of the "feature fusion + single classifier" paradigm used in EFM-Net. Although EFM-Net captures key part information of ships via its attention module, it compresses all features into a single representation, which fails to fully leverage the complementary information between shallow edge features and deep semantics features. In contrast, the superior performance of our SHKF-PG model can be primarily attributed to its novel "SHKF + multiexpert ensemble" framework. Our SHKF-PG model simultaneously extracts shallow local features,

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middle-layer attention features, and deep global features through a multilevel feature extraction module. The SHKF then assigns distinct feature combinations to different experts, enabling each expert to specifically mine discriminative information from unique perspectives. Furthermore, the PG mechanism effectively suppresses consensus errors among experts on nonship negative classes or similar ship negative classes, thereby mitigating decision biases caused by background interference. Consequently, this experiment validates the superiority of our model in complex scenes with large intraclass differences.

2) Performance on FGSCR-42: This experiment evaluates the model's discriminative capability for ship subcategories with extremely subtle interclass differences. The FGSCR-42 dataset includes more fine-grained categories with minimal interclass variation, demanding greater ability to capture subtle discriminative features. The results in Table II show that our model also achieves the highest OA on the FGSCR-42 dataset among all compared methods. Compared to the latest MFAL, Cog-Net, and MGFE-Net models, the SHKF-PG model improves the performance by 1.8%, 1.3%, and 0.4%, respectively. While Cog-Net extracts both global and local features, it often confuses the hardest negative classes due to insufficient feature discriminability. In contrast, the PG mechanism in SHKF-PG accurately identifies the consensus tendency among experts toward the hardest negative classes and guides them to strengthen learning of subtle differentiating features. This result confirms that our model exhibits stronger discriminative abilities in fine-grained classification with high interclass similarity and low intraclass similarity.

3) Performance on Aircraft-16: This experiment assesses the model's fine-grained classification capability for aircraft targets. The results in Table III show that our model achieves an OA of 92.6% on the Aircraft-16 dataset. It outperforms EFM-Net (OA = 92.0%) by 0.6%, which is designed for remote sensing feature extraction. Compared to the SIM and SGFM-Net model, our model shows a substantial improvement of 3.1% and 3.3%, respectively. The improvements demonstrate that our SHKF-PG model, with its multilevel feature extraction and multiexpert decision framework, generalizes well across different target types and can be effectively adapted to various fine-grained object classification tasks in remote sensing images.

D. Ablation Study

In this section, we design ablation studies to validate the effectiveness of SHKF-PG and its core components.

1) Effectiveness of the Proposed Modules: In this experiment, taking EFM-Net as the baseline, we progressively integrate the SHKF network, MKD among experts, and the PG mechanism into the SHKF-PG model to analyze the individual and combined contributions of each module. The results of incrementally adding these components are presented in Table IV.

As shown in Table IV, we can find that introducing the SHKF network alone brings a significant performance gain (+0.6% OA), underscoring the effectiveness of the core "self-heterogeneous fusion" design. This demonstrates that simply by replacing the standard feature fusion with our specialized fusion and expert assignment strategy, the model's capacity to utilize multilevel features is substantially enhanced, laying a solid foundation for the multiexpert ensemble. When both SHKF and MKD are applied, performance further rises to 92.6%, indicating that MKD facilitates collaboration among experts through message exchange and reduces individual cognitive bias. Furthermore, the PG mechanism accurately suppresses the consensus negative class by constructing tailored guidance signals, thereby preventing the propagation of wrong consensus during knowledge transfer. When all three modules are integrated, the model achieves the highest OA of 93.0%, demonstrating their complementary roles in enhancing model performance.

TABLE IV
ABLATION STUDIES OF THE PROPOSED MODULES IN THE SHKF-PG MODEL ON THE FGSC-23 DATASET

#	SHKF	MKD	PG	OA(%)
1	×	×	×	91.6
2	✓	×	×	92.2 (+0.6)
3	✓	✓	×	92.6 (+1.0)
4	✓	×	✓	92.7 (+1.1)
5	✓	✓	✓	**93.0 (+1.4)**

TABLE V
ABLATION STUDIES OF THE NUMBER OF EXPERTS AND VARIANTS ON THE FGSC-23 DATASET

Expert Num.	Variants	Expert feature sources	OA
1	Assignment Rule	F^M	91.6
2	Assignment Rule	F^M,F_3^L	92.5
	✓	F^M,F_1^L	92.1
	✓	F^M,F_2^L	92.3
3	Assignment Rule	F^M,F_2^L,F_3^L	**93.0**
	✓	F^M,F_1^L,F_2^L	92.7
	✓	F^M,F_1^L,F_3^L	92.5
4	Assignment Rule	F^M,F_1^L,F_2^L,F_3^L	92.6

2) Exploring the Impact of the Number of Experts: To balance discriminative power and model complexity, we design the ablation study to determine the optimal number of experts and their feature combination variants. In this experiment, we assigned distinct feature combinations to experts in quantities of 2 and 3. For instance, in a two-expert configuration, inputs consist of middle-layer attention features and distinct shallow local features. Performance under various expert counts and feature combinations is summarized in Table V.

From Table V, both single-expert and two-expert configurations underperform compared to the three-expert setup. With too few experts, the model fails to make full use of different hierarchical features. For example, the two-expert model using F^M and F_1^L attains only 92.1% OA, due to the absence of shallow texture information from F_2^L. Notably, among all two-expert variants, the combination prescribed by our assignment rule (F^M and F_3^L) achieves the highest accuracy (92.5%),

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TABLE VI
ABLATION STUDIES OF THE DIFFERENT PARAMETERS ON THE FGSC-23 DATASET

α	0	0.1	0.2	0.3	0.1	0.1	0.1	0
β	0	0	0	0	0.1	0.2	0.3	0.2
OA (%)	92.2	92.6	92.4	91.9	92.7	**93.0**	92.7	92.5

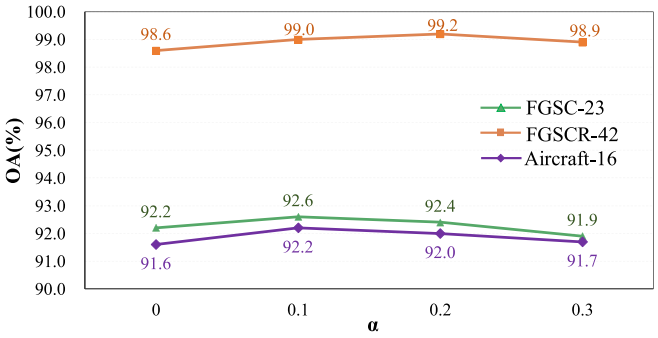


Fig. 5. Sensitivity analysis of hyperparameter α.

demonstrating that our systematic allocation of deeper shallow features yields better feature complementarity. When three experts were employed, combining mid-level attention features (F^M) with shallow local features F_2^L and F_3^L yielded an optimal OA of 93.0% across all combinations. Increasing the number of experts to four actually reduced the model's OA to 92.6%. It indicates excessive experts introduce redundant features, increase computational cost, and complicate optimization. Thus, we adopt a three-expert configuration in SHKF-PG.

3) Impact of Different Hyperparameters on Model Performance: The SHKF-PG model involves two key hyperparameters: the weight of MKD α and the weight of PG loss β. We perform a sensitivity analysis to identify values that balance expert collaboration and PG. Model performance under different hyperparameter settings is shown in Table VI.

The parameter α controls the degree of knowledge transfer among experts. As indicated in Table VI, setting α too high (e.g., $\alpha = 0.3$) reduces OA to 91.9%, which is 0.7% lower than with $\alpha = 0.1$. The declines indicate that excessive knowledge distillation forces experts toward similar distributions, diminishing their ability to capture distinctive features and reducing classification accuracy. In addition, we also tested the parameter sensitivity of α on FGSCR-42 and Aircraft-16 datasets, as shown in Fig. 5. We consistently observed that when $\alpha \leq 0.2$, accuracy improves compared to $\alpha = 0$, with gains ranging from 0.4% to 0.6% across datasets. When $\alpha \geq 0.3$, performance begins to degrade. For adaptation to new datasets, it is recommended to tune α within the range from 0 to 0.3.

The parameter β modulates the strength of PG. When $\beta = 0.1$, the model begins to effectively suppress hard negative classes, improving model performance by 0.1% (92.7% versus 92.6%) compared to the setup without guidance ($\beta = 0$). When β is excessively large (e.g., $\beta = 0.3$), overly strong PG undermines the experts' own learning ability. With $\alpha = 0.1$ and $\beta = 0.2$, the model achieves 93.0% performance. Consequently, $\alpha = 0.1$ and $\beta = 0.2$ are identified as the optimal hyperparameter combination on the FGSC-23 dataset.

4) Variants of the SHKF: To further investigate the effect of the SHKF network, we design a variant in the experiments. For self-heterogeneous multiexperts, shallow local features from different stages are fed directly into expert classifiers without fusion with deep global features. Experiments are conducted on all three datasets, with results provided in Table VII.

As shown in Table VII, even without deep feature fusion, the multiexpert model improves OA by 0.8% over the baseline (from 91.6% to 92.4%), indicating that the multiexpert framework itself enhances the utilization of multilevel features. After incorporating deep global feature fusion via SHKF, its performance is further improved across all datasets: OA increases from 92.4% to 93.0% on FGSC-23, from 99.2% to 99.4% on FGSCR-42, and from 92.2% to 92.6% on Aircraft-16. These results demonstrate that self-heterogeneous feature fusion equips multiple experts with stronger discriminative capabilities, enabling them to narrow down candidate decisions using global semantics while focusing on discriminative local details for final identification. Therefore, we adopt the strategy of self-heterogeneous feature fusion combined with multiexpert classification, which effectively leverages both global and local information to enhance discriminative power.

5) Variants of the PG Mechanism: To validate the superiority of the proposed PG mechanism in suppressing consensus on negative classes, we compare it against three alternative strategies: a baseline without any guidance (w/o), a cosine diversity loss, and the Nt loss proposed in [13] for hardest negative class suppression in long-tailed recognition. Results are summarized in Table VIII.

The cosine diversity loss encourages divergence among expert predictions to increase diversity. As shown in Table VIII, after incorporating the cosine diversity loss, the model's performance improved by 0.1% compared to the baseline. It suggests that increasing diversity can indirectly mitigate consensus on hard negatives. However, it lacks explicit guidance toward correct classifications. Besides, equipping the model with the Nt Loss leads to a performance drop of 0.3% compared to the baseline, indicating that methods designed for long-tailed recognition are not directly applicable to remote sensing fine-grained classification. The PG mechanism is a crucial component designed to maintain the health of the "multiexpert ensemble." It indirectly guides the model to reduce confidence in the incorrect class, rather than directly modifying the output logits. While the framework encourages diversity among experts, it is critical to prevent them from forming a consensus on incorrect predictions. As shown in Table VIII, our PG loss successfully fulfills this role, outperforming the baseline and the cosine diversity loss by 0.4% and 0.3%, respectively. It ensures that the valuable diversity introduced by the SHKF is guided productively toward the correct decision, rather than diverging into error.

6) Analysis of Individual Expert Performance: We further analyze the performance of individual experts and their ensembles to verify complementarity among experts specializing in different feature hierarchies. The OA results of each expert independently and under different ensemble combinations are

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TABLE VII
VARIANTS STUDIES OF THE SHKF

Method	Input Features	FGSC-23 (OA%)	FGSCR-42 (OA%)	Aircraft-16 (OA%)
Baseline(w/o SHKF)	Single global representation	91.6	99.2	92.0
Self-heterogeneous multi-experts	Different level features	92.4	99.2	92.2
Self-heterogeneous knowledge fusion	Self-heterogeneous fused features	**93.0**	**99.4**	**92.6**

TABLE VIII
VARIANTS STUDIES OF THE PG LOSS. "W/O" DENOTES MODEL WITHOUT PG

Method	Principle	FGSC-23 (OA%)	FGSCR-42 (OA%)	Aircraft-16 (OA%)
Baseline (w/o PG)	-	92.6	99.2	92.3
+Cosine penalty loss	Increases prediction diversity	92.7	99.3	92.4
+NT loss	Suppresses hardest negative class (for long-tailed)	92.3	99.1	92.1
+PG loss	Dynamically suppresses consensus negative class	**93.0**	**99.4**	**92.6**

TABLE IX
ANALYZING THE PERFORMANCE OF INDIVIDUAL EXPERT AND THEIR COMBINATIONS ON THE FGSC-23 DATASET

Expert Combination	Input Features	OA (%)	Improvements
Expert 1	$F_1^D + F_3^L$	87.5	-
Expert 2	$F_2^D + F_2^L$	90.1	-
Expert 3	$F_3^D + F^M$	90.8	-
Ensemble (1+2)	-	91.6	+1.5 (vs. E2)
Ensemble (1+3)	-	92.1	+1.3 (vs. E3)
Ensemble (2+3)	-	91.4	+0.6 (vs. E3)
Full Ensemble (1+2+3)	-	**93.0**	**+2.2 (vs. E3)**

TABLE X
PERFORMANCE WITH DIFFERENT EXPERT ENSEMBLE STRATEGIES ON THE FGSC-23 DATASET

Ensemble strategies	Description	OA
Weighted by Confidence	Weights based on experts' max score	92.3
Task-based Weighting	Fixed weights: $[0.25, 0.25, 0.5]$	92.5
Gated Weighting	Weights generated by a gate unit	92.7
Sum Averaging	Equal weights for all experts	**93.0**
SF-Expert Gated Network	Each expert receives a single feature	88.6
Feature-Gated Network	A gating network to select features	92.0

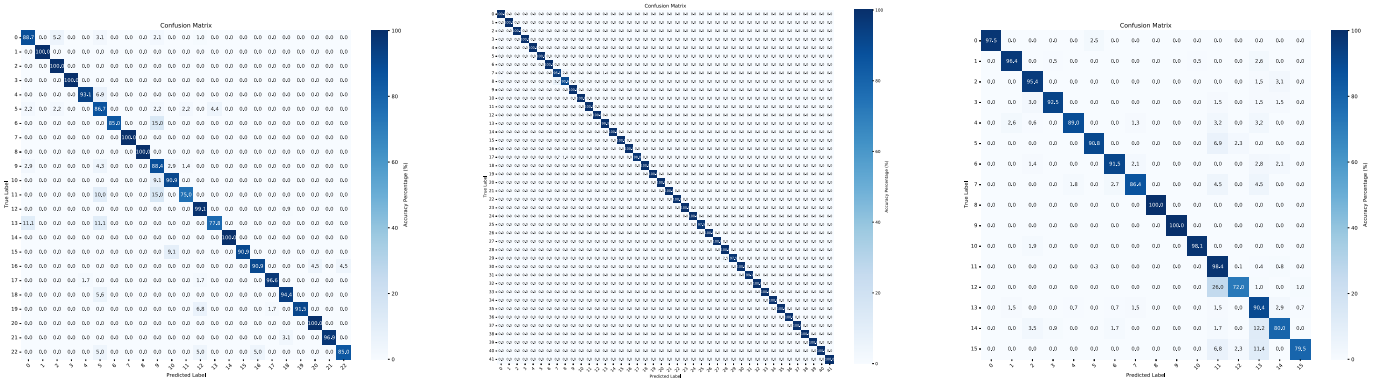
presented in Table IX. As shown in Table IX, we can observe that the performance of individual experts underperforms compared to their ensembles. Specifically, Expert 1, which takes shallow local features and deep global features as input, achieves an OA of only 87.5%. Expert 3, which utilizes middle attention features and deep global features, performs better with an OA of 90.8%. Ensembling any two experts consistently improves performance. For instance, combining Experts 1 and 3 raises OA by 4.6% and 1.3% relative to their individual results. Besides, the full ensemble of all three experts achieves the highest OA of 93.0%, significantly outperforming all partial combinations. The improvements indicate that each expert in our SHKF-PG can focus on mining discriminative information from the feature dimension it specializes in, maximizing the value of each feature and ultimately achieving efficient complementarity at the decision level. The ensemble effectively integrates information from shallow edge features (Expert 1), shallow local features (Expert 2), and mid-level part attention (Expert 3), compensating for individual limitations and enhancing overall discriminability.

7) Impact of Different Expert Ensemble Strategies: In this experiment, we evaluate four ensemble strategies for combining expert outputs: weighted by confidence, task-based weighting, gated weighting, and sum averaging. The performance results with different expert ensemble strategies on FGSC-23 dataset are presented in Table X.

The weighted confidence strategy weights based on experts' max softmax score. The task-based weighting strategy assigns fixed weights to different experts for integration. Empirically, we assign weights of 0.25, 0.25, and 0.5 to the three experts. The gated weighting strategy employs a trainable gate unit to dynamically weight experts. The sum averaging strategy treats the decision of each expert equally and aggregates their outputs by averaging. As shown in Table X, sum averaging achieves the highest OA, outperforming task-based and gated weighting by 0.5% and 0.3%, respectively. The main reason is that under limited training data where some categories have few samples and high interclass similarity, the gating network is prone to sample bias and overfitting. In contrast, sum averaging without introducing bias or additional parameters avoids overfitting and fully leverages the complementary capabilities of different experts. We also conduct additional experiments with several gating-based variants. The SF-Expert gated network achieves only 88.6% OA, which indicates that isolated features lack sufficient discriminative power for fine-grained classification. The feature-gated network achieves 92.0% OA, but still underperforms our SHKF-PG. Therefore, we adopt sum averaging as the default ensemble strategy in SHKF-PG.

8) Impact of Different Feature Fusion Strategies: To validate the rationality of selecting the Hadamard product for feature fusion in (5)–(8), we designed ablation experiments to compare it with two widely used fusion strategies: Addition and Concatenation. The performance results with different

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(a) (b) (c)

Fig. 6. Confusion matrices of SHKF-PG model on three datasets. (a) FGSC-23, (b) FGSCR-42, and (c) Aircraft-16. The number in the square on the diagonal of the matrix represents the AR of each category.

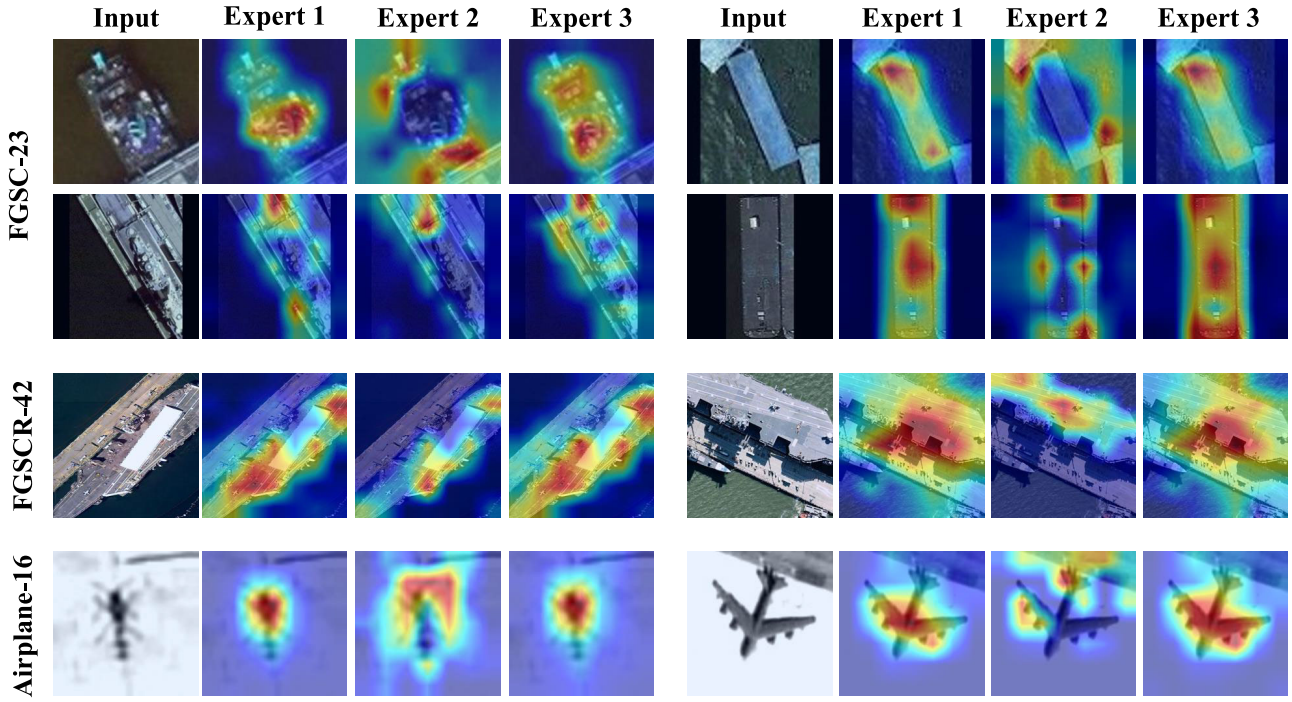


Fig. 7. Difference between three experts on feature activation maps, which are generated by Grad-CAM.

TABLE XI

EFFECT OF DIFFERENT FEATURE FUSION STRATEGIES FOR SHKF-PG

Feature fusion strategies	OA
Addition	92.0
Concatenation	92.8
Hadamard product	**93.0**

TABLE XII

COMPUTATIONAL EFFICIENCY RESULTS OF SHKF-PG AND EFM-NET MODEL

Model	Params	Gflops	FPS	Training Time	Test Speed	TVM	Acc
EFM-Net w/o Refiner*	88.0M	15.36	215.53	0h58min	0.0880s/item	10070M	91.6
EFM-Net*	88.0M	15.36	208.25	3h58min	0.1471s/item	13723M	92.3
Ours	147.3M	18.36	170.95	1h09min	0.0964s/item	11012M	**93.0**

feature fusion strategies on FGSC-23 dataset are presented in Table XI.

From Table XI, we can find that simply summing the features tends to dilute their distinct semantics, and it only achieves 92.0% OA. Concatenation preserves information better and improves accuracy, but it often increases computational complexity and leads to information redundancy. The Hadamard product achieves the best performance among the three strategies. Its computational cost is comparable to that of summation, making it both effective and efficient.

E. Discussion

1) Analysis of Model Complexity: Table XII presents a comparative analysis of the computational efficiency among

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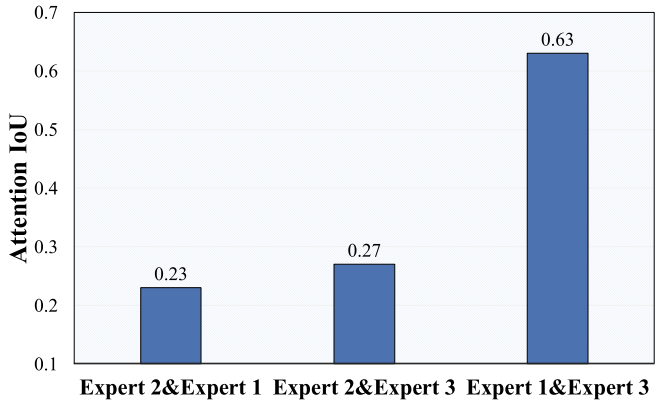


Fig. 8. Attention IoU between three experts on the FGSC-23 dataset.

EFM-Net w/o Refiner, EFM-Net, and SHKF-PG, including model parameters (Params), computational cost (FLOPs), training efficiency (training time), inference efficiency (FPS and test speed), training video memory (TVM), and accuracy (Acc). All experiments were conducted under identical hardware and software conditions to ensure fairness: an NVIDIA GeForce 3090Ti GPU with 24-GB memory, with a batch size of 12 and input resolution 224×224, and using FP32 precision.

SHKF-PG exhibits a moderate increase in parameters compared to EFM-Net, which is primarily attributed to tripling the expert classifiers. Notably, while our multiexpert structure triples the number of classifiers, the total parameter count increases by only 67.4%, as the increase stems mainly from the dedicated parameters of each expert.

Despite the increase in parameters, both the training and inference efficiency of our model are significantly superior to those of EFM-Net. Efficient training can substantially reduce research and application costs. The results indicate that the training time of our model is about 1 h, which is approximately 71.2% shorter than that of EFM-Net, representing a remarkable improvement in training efficiency. It is important to note that the Refiner module in EFM-Net is different from conventional data augmentation. During each iteration, the Refiner identifies discriminative regions from attention maps, generates two additional augmented samples, and then processes all three samples through the network, requiring three separate forward and backward passes. This inherent design triples the computational cost per batch and is the primary reason why EFM-Net's training time is substantially longer than our model's. In summary, with a modest increase in resource consumption, our model achieves higher accuracy and, importantly, much greater efficiency than the previous models, resulting in a better overall tradeoff.

2) Analysis of the AR: To fully demonstrate the advantages of our SHKF-PG model, we visualize the confusion matrices for all three datasets. Fig. 6 presents the confusion matrices for FGSC-23, FGSCR-42, and Aircraft-16, respectively. In Fig. 6, the horizontal axis represents predicted labels, while the vertical axis denotes true labels. Besides, the values on the diagonal indicate the AR for each category.

On the FGSC-23 dataset, Class 13 is easily confused with Classes 0, 5, 9, and 11. However, as illustrated in Fig. 6(a), our model can distinguish Class 13 from Classes 9 and 11 by leveraging multiexpert collaboration to fully utilize features from different levels. Overall, we observe that 7 classes achieve an AR of 100%, and 14 classes exceed 90% AR on the FGSC-23 dataset. On the FGSCR-42 dataset, an impressive 35 classes achieve perfect AR. For Aircraft-16, 11 classes attain an AR above 90%. These results confirm that SHKF-PG excels at discriminating between visually similar categories.

3) Qualitative Analysis: In this section, we visualize the class activation maps and prediction results of different experts. As shown in Fig. 7, we employ Grad-CAM [63] to generate the feature activation maps for each expert. In the activation maps, red and blue regions indicate our model's attention and suppression, respectively.

It can be observed that the three experts exhibit distinct attention patterns. Expert 1's attention regions typically cluster around objects or areas with sharp edge variations, indicating its focus on valuable shallow edge information. The attention of Expert 2 is usually concentrated on localized object areas, covering relatively small regions, suggesting its primary focus on discriminative shallow parts or structures. Expert 3's attention regions are generally larger, with scattered and loosely defined focus areas. Furthermore, although Expert 1 alone achieves the lowest OA (87.5%, Table IX), its integration significantly enhances ensemble performance, demonstrating its ability to complement other experts by balancing global and local information. To further validate the specialization of each expert, we computed the intersection over union (IoU) of attention maps [64] on the FGSC-23 dataset. The average IoU values are shown in Fig. 8.

From Fig. 8, we can find that Expert 2 exhibits substantial differences from both Experts 1 and 3, with IoU values of 0.23 and 0.27, respectively. It confirms that Expert 2 focuses on regions distinct from those attended by Experts 1 and 3, supporting our claim that it specializes in edge-level information. Besides, Experts 1 and 3 also show noticeable differences. Expert 1 focuses on local regions through shallow local features, while Expert 3 attends to discriminative parts via mid-level attention features. It is reasonable that they share some overlap in their attention regions. However, the IoU value of 0.63 indicates that they are far from identical, highlighting their complementary expertise.

Fig. 9 presents the prediction probabilities for individual experts and their ensemble. For clarity, we mainly show the top five predicted categories and their corresponding probabilities for each expert. It is evident that for simpler samples, the three experts yield consistent predictions. But for challenging or ambiguous samples, the three experts make slightly divergent judgments based on feature inputs from different layers. The ensemble mechanism synthesizes these diverse predictions, resulting in robust and accurate decisions.

In summary, the visualizations of both the feature activation maps and prediction results demonstrate that valuable information from different hierarchical features is effectively learned by different experts through the SHKF network. In decision-

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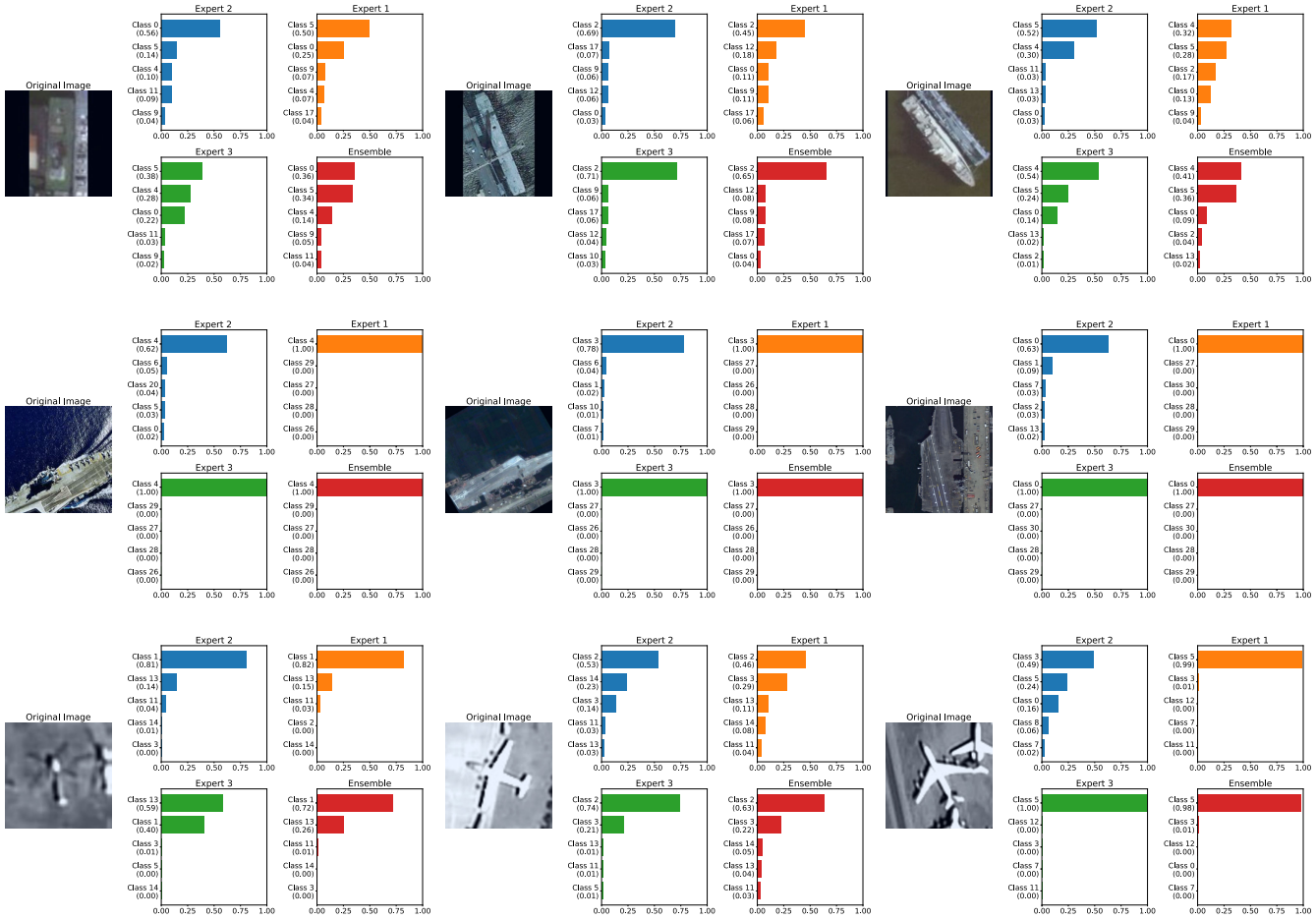


Fig. 9. Prediction results of different experts.

making, their complementary information enhances the overall performance of SHKF-PG.

V. Conclusion

In this article, we propose a novel SHKF-PG model to address the limitations of conventional "feature fusion + single classifier" paradigms in FGVC-RS. The SHKF-PG introduces a "SHKF + multiexpert ensemble" framework that effectively enhances the utilization of discriminative information while reducing information loss. The SHKF network assigns distinct hierarchical features to dedicated experts for valuable information extraction, which significantly improves feature utilization. The PG mechanism dynamically handles hard samples by suppressing consensus on the hardest negative class among experts, steering the model toward the correct optimization direction. Extensive experimental results on three public benchmarks demonstrate that our model achieves SOTA performance. Future work will incorporate soft label assignment based on semantic similarity and explore long-tailed fine-grained classification.

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